UNIT 1 · 9 hours

Analysis of Pure R, L and C Circuits



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

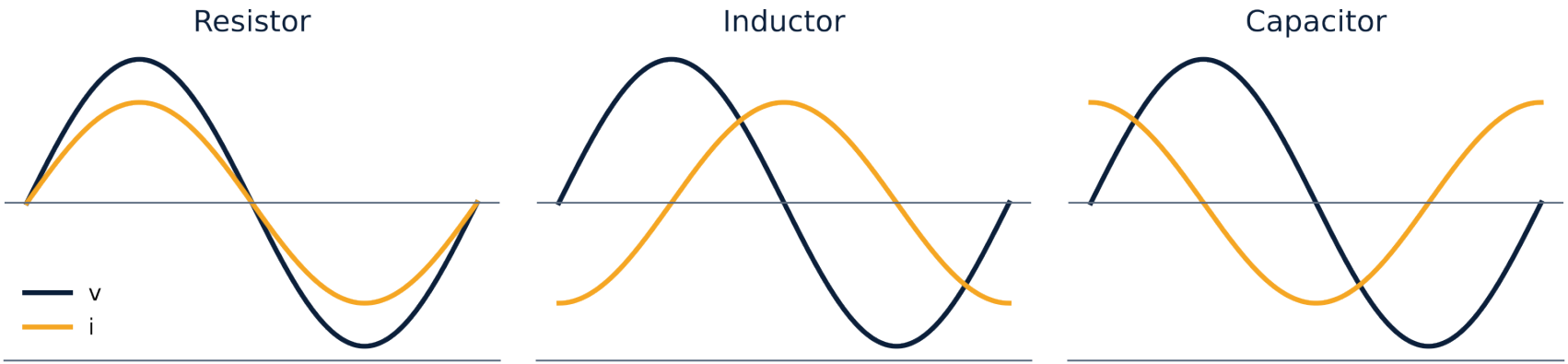
- Derive current and power in pure R, L and C
- Define inductive and capacitive reactance
- Draw phasor diagrams and waveforms
- Solve numericals on each element

Syllabus context

Analysis of R, L, C, RL, RC and RLC series circuits, real power, reactive power, apparent power and power factor.

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Overview: v and i in R, L and C



- R: i in phase with v.
- L: i lags v by 90°.
- C: i leads v by 90°.
- Power: R consumes, L and C only store and return energy.

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Pure resistor

- $v = V_m \sin \omega t$; $i = v/R = I_m \sin \omega t$, $I_m = V_m/R$
- Current and voltage in phase ($\varphi = 0°$).
- $p = vi = V_m I_m \sin^2 \omega t = (V_m I_m/2)(1 - \cos 2\omega t)$
- Average power $P = V_m I_m/2 = V \cdot I$ (RMS values)
- Power factor = 1

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Pure inductor

- $v = L\, di/dt$, so $i = (1/L)\int v\, dt = I_m \sin(\omega t - 90°)$
- Inductive reactance $X_L = \omega L = 2\pi f L$ (Ω); $I_m = V_m / X_L$
- Current lags voltage by 90°.
- $p = -(V_m I_m/2) \sin 2\omega t$: average power is zero.
- Energy $\tfrac{1}{2}LI^2$ is stored in the magnetic field and returned each quarter cycle.

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Pure capacitor

- $i = C\, dv/dt = \omega C\, V_m \cos \omega t = I_m \sin(\omega t + 90°)$
- Capacitive reactance $X_C = 1/(\omega C) = 1/(2\pi f C)$ (Ω)
- Current leads voltage by 90°.
- Average power is zero; energy $\frac{1}{2}CV^2$ is stored in the electric field.
- X_C falls with frequency: at DC a capacitor is an open circuit.

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Comparison of the three elements

	R	L	C
Opposition	R (Ω)	XL = 2πfL	XC = 1/2πfC
Phase of i w.r.t. v	0°	−90° (lags)	+90° (leads)
Average power	VI	0	0
Energy	dissipated	magnetic field	electric field
Effect of f	none	XL ∝ f	XC ∝ 1/f

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Numerical: inductor

Problem A pure inductor takes 20 A from a 230 V, 50 Hz supply. Find XL, L, power and the equations of v and i.

- $X_L = V/I = 230/20 = 11.5\ \Omega$
- $L = X_L/(2\pi f) = 11.5/314 = 0.0366$ H
- $V_m = 325.27$ V, $I_m = 28.28$ A, $\omega = 314$
- $P = 0$

Answer: $v = 325.27 \sin 314t$ V, $i = 28.28 \sin(314t - 90^\circ)$ A

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Numerical: capacitor

Problem A 400 µF capacitor is connected to 230 V, 50 Hz. Find XC, the RMS current and the equations.

- $X_C = 1/(2\pi \cdot 50 \cdot 400 \times 10^{-6}) = 7.95\ \Omega$
- $I = V/X_C = 230/7.95 = 28.93$ A; $I_m = 40.91$ A

Answer: $i = 40.91 \sin(314t + 90°)$ A, $P = 0$

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Key takeaways

- R: in phase; L: current lags 90°; C: current leads 90°
- $X_L = 2\pi fL$, $X_C = 1/(2\pi fC)$
- Average power in pure L or C is zero
- Reactive elements store and return energy

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Practice questions

Q1. A 50 Ω resistor is on 230 V, 50 Hz. Find I, P and write i(t).

Q2. Find the reactance of a 0.1 H inductor at 50 Hz and 500 Hz.

Q3. A 100 µF capacitor on 230 V, 50 Hz: find current and reactive power.

Q4. At what frequency does a 10 mH inductor have a reactance of 100 Ω?